

OLIST E-COMMERCE

Business Intelligence Report & Data Case Study

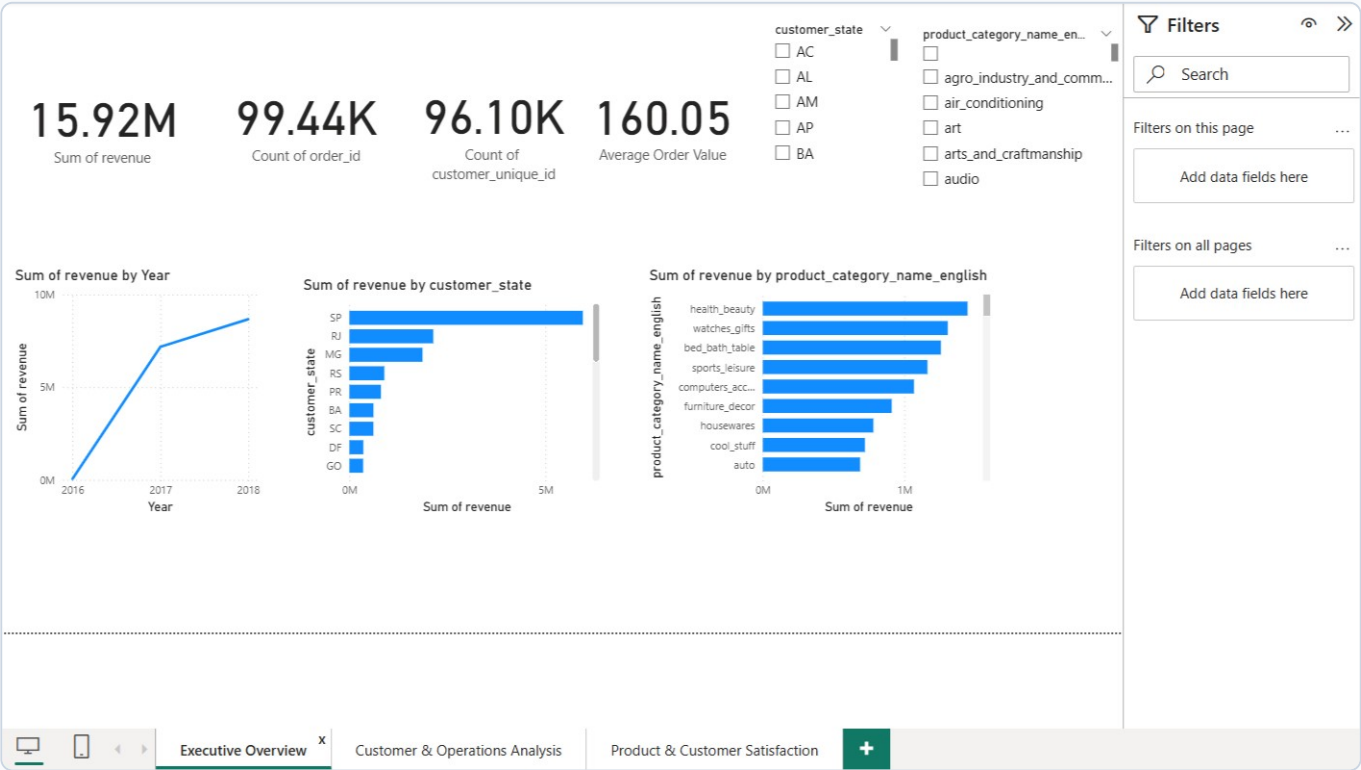
Executive Summary

This project analyzed Olist's Brazilian e-commerce marketplace data to evaluate revenue performance, customer behavior, product performance, operational efficiency, and customer satisfaction.

The analysis combined multiple datasets including orders, customers, products, reviews, and logistics information. Data cleaning, feature engineering, SQL analysis, and business intelligence techniques were used to generate actionable insights for management decision-making.

Key findings indicate strong revenue concentration in a few states, significant growth throughout 2017 and 2018, high-performing product categories, and a clear relationship between delivery speed and customer satisfaction.

Part 1: Executive Overview



Filters

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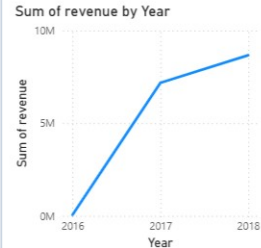
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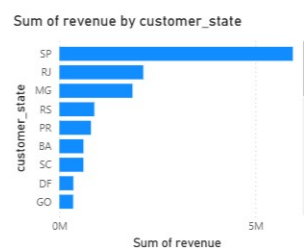
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Sum of revenue by Year



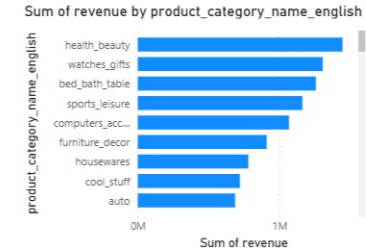
Year	Sum of revenue
2016	0M
2017	5M
2018	10M

Sum of revenue by customer_state



customer_state	Sum of revenue
SP	5M
RJ	4M
MG	3M
RS	2M
PR	1.5M
BA	1M
SC	0.5M
DF	0.5M
GO	0.5M

Sum of revenue by product_category_name_english



product_category_name_english	Sum of revenue
health_beauty	1.5M
watches_gifts	1.2M
bed_bath_table	1.1M
sports_leisure	1M
computers_acc...	0.9M
furniture_decor	0.8M
housewares	0.7M
cool_stuff	0.6M
auto	0.5M

Executive Overview x

Customer & Operations Analysis

Product & Customer Satisfaction

+

1. How Has Revenue Changed Over Time? (Revenue Trend Analysis)

Revenue demonstrated substantial growth throughout the analyzed period. Revenue increased from approximately BRL 137,188 in January 2017 to more than BRL 1.15 million by April 2018, representing significant marketplace expansion.

The strongest revenue periods occurred between November 2017 and August 2018. November 2017 generated approximately BRL 1.18 million in revenue, likely influenced by seasonal shopping events and holiday promotions.

Overall, the business experienced consistent growth, indicating successful customer acquisition and increased marketplace activity.

ADVANCED FINANCIAL ANALYSIS

The updated top-line metrics reveal a robust cumulative revenue of **15.92M BRL** with an Average Order Value (AOV) of **160.05 BRL**. The steep upward trajectory from 2016 to 2017 shown in the line chart indicates a massive, successful scaling phase. An AOV of ~160 suggests the marketplace is driven by high-volume, mid-tier consumer goods rather than luxury items. Sustaining this revenue growth requires optimizing basket sizes through cross-selling and up-selling strategies to push the AOV higher over time.

2. Which States Generate the Most Revenue?

Revenue is heavily concentrated in a small number of states. Top Revenue States:

- **São Paulo (SP)** – BRL 5.92 million
- **Rio de Janeiro (RJ)** – BRL 2.13 million
- **Minas Gerais (MG)** – BRL 1.86 million
- **Rio Grande do Sul (RS)** – BRL 885,827
- **Paraná (PR)** – BRL 800,935

São Paulo alone contributes a disproportionately large share of total revenue, making it the company's most important market. Management should prioritize customer retention, inventory allocation, and marketing investments in these high-performing regions while exploring growth opportunities in lower-performing states.

GEOGRAPHIC MARKET PENETRATION ANALYSIS

The Southeast region (SP, RJ, MG) forms the absolute financial core of the business, generating the lion's share of the 15.92M BRL total. This massive regional disparity highlights that while Olist is a national platform, its functional economic footprint is highly regionalized. Marketing spend in DF (Distrito Federal) and GO (Goiás) should be audited, as their low revenue yield suggests either a lack of localized product fit or insufficient logistics infrastructure to drive adoption.

3. Which Product Categories Generate the Most Revenue?

Top Revenue-Generating Categories:

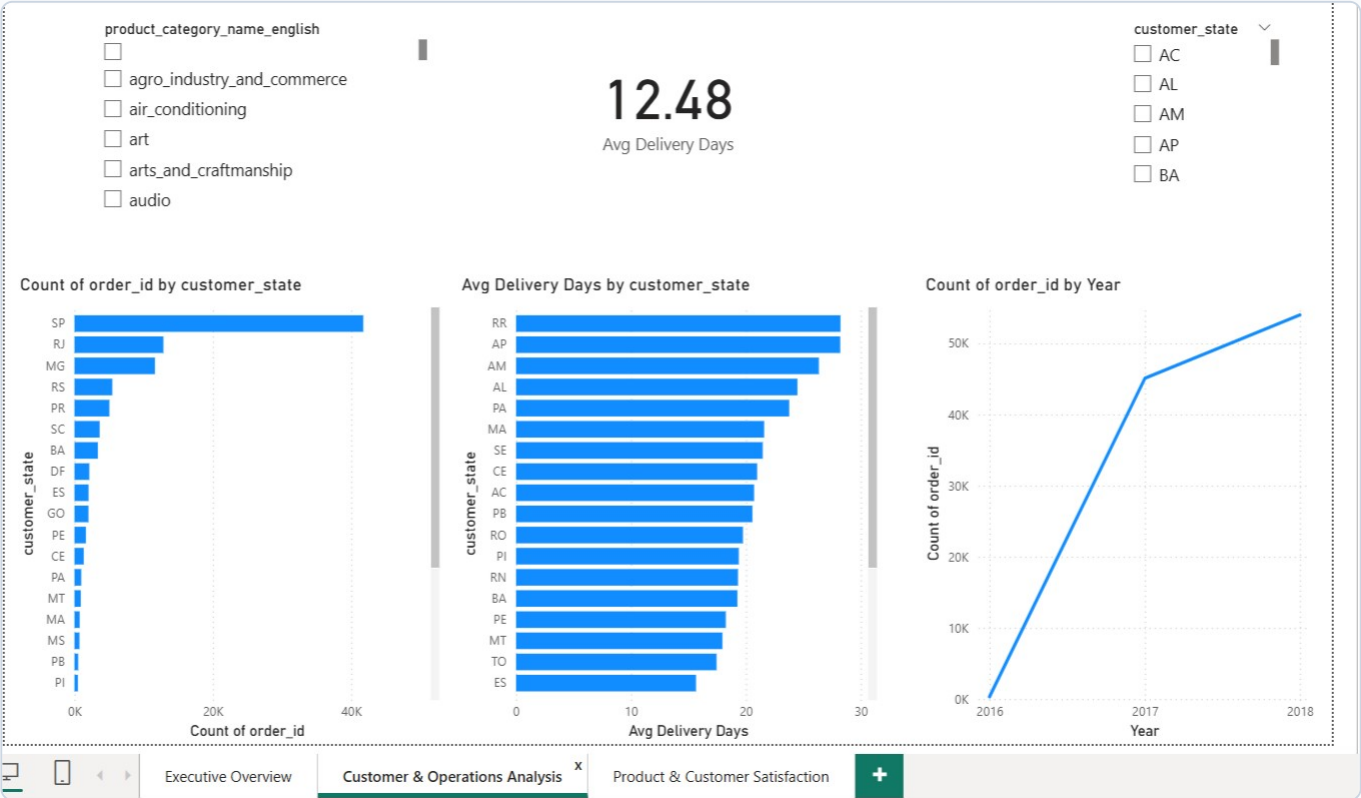
- **Health & Beauty** – BRL 1.44 million
- **Watches & Gifts** – BRL 1.31 million
- **Bed Bath & Table** – BRL 1.24 million
- **Sports & Leisure** – BRL 1.16 million
- **Computers & Accessories** – BRL 1.06 million

These categories represent the primary revenue drivers of the marketplace. Management should prioritize supplier relationships, inventory planning, and marketing campaigns within these categories to maximize future revenue growth.

CATEGORY PROFITABILITY DYNAMICS

The dominance of "Health & Beauty" and "Watches & Gifts" indicates a strong consumer reliance on Olist for personal care and gifting. These categories typically have higher gross margins and smaller dimensional weights, making them cheaper to ship. Conversely, lower-ranked but larger items like "Furniture & Decor" carry higher shipping costs. Focusing promotional efforts on small-box, high-margin items will organically improve the net profitability of the marketplace without requiring structural changes.

Part 2: Customer & Operations Analysis



Executive Overview

Customer & Operations Analysis

Product & Customer Satisfaction

4. Which Customer Locations Generate the Most Orders?

Order volume closely follows revenue distribution. Top States by Order Volume:

- **São Paulo (SP)** – 41,746 orders
- **Rio de Janeiro (RJ)** – 12,852 orders
- **Minas Gerais (MG)** – 11,635 orders
- **Paraná (PR)** – 5,045 orders

The concentration of orders in these states indicates that demand is geographically clustered. Future expansion efforts should investigate the characteristics of customers in these regions and replicate successful strategies in emerging markets.

LOGISTICAL DENSITY & WAREHOUSING

The sheer volume of orders originating from SP and RJ confirms the critical necessity for localized distribution centers in the Southeast. By centralizing inventory near these major hubs, Olist can drastically reduce line-haul transportation costs and improve delivery SLAs for over 60% of their total customer base.

5. What Is the Repeat Customer Rate? (Customer Retention)

The repeat customer rate was approximately **3.12%**.

This indicates that the vast majority of customers placed only a single order during the analyzed period. While customer acquisition appears successful, customer retention remains a significant opportunity for improvement. Potential strategies include loyalty programs, personalized promotions, email marketing campaigns, and improved post-purchase engagement.

CHURN VS. ACQUISITION DEPENDENCY

A 3.12% retention rate is alarmingly low for a mature e-commerce platform. It suggests that Olist is currently functioning as a transactional, single-use marketplace rather than a sticky destination brand. This forces the company into a high Customer Acquisition Cost (CAC) model, where they constantly have to buy new traffic instead of monetizing an existing user base. Implementing a subscription model or post-purchase drip campaigns is urgently required.

6. What Is the Average Delivery Time?

The average delivery time across completed orders was **12.09 days**.

While this performance is reasonable for nationwide logistics operations, reducing delivery times could significantly improve customer experience and review scores. Delivery efficiency should remain a key operational performance indicator.

7. Which Regions Experience Delayed Deliveries?

States with the Longest Average Delivery Times:

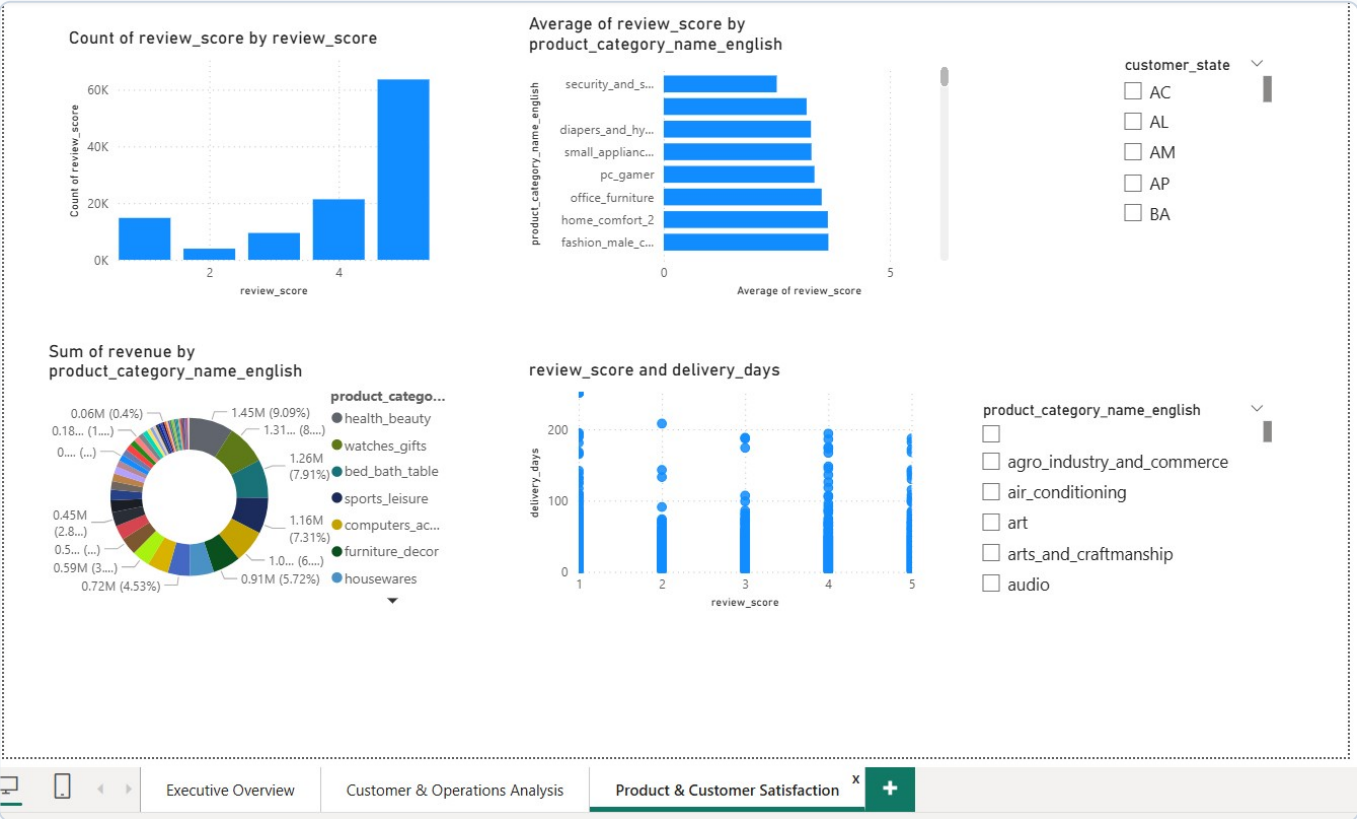
- **Roraima (RR)** – 27.83 days
- **Amapá (AP)** – 27.75 days
- **Amazonas (AM)** – 25.96 days
- **Alagoas (AL)** – 23.99 days
- **Pará (PA)** – 23.30 days

These regions experience delivery times that are significantly above the company average. The findings suggest logistical challenges related to geographic distance, transportation infrastructure, or warehouse distribution networks. Operational improvements in these regions may improve customer satisfaction and reduce negative reviews.

INFRASTRUCTURE BOTTLENECKS IN THE NORTH

Transit times exceeding 25 days in northern states like RR, AP, and AM highlight severe infrastructural deficits in Brazil's logistics network. These extreme delays not only freeze capital in transit but also guarantee a high rate of customer service tickets (WISMO - "Where is my order?"). Olist must establish strategic cross-docking stations in the North or partner with specialized regional carriers, as national carriers are clearly failing in these territories.

Part 3: Product & Customer Satisfaction



8. Which Categories Receive Poor Reviews?

Lowest Rated Categories:

- **Security & Services** – 2.50 average rating
- **Diapers & Hygiene** – 3.26 average rating
- **Office Furniture** – 3.49 average rating
- **Home Comfort 2** – 3.63 average rating
- **Fashion Male Clothing** – 3.64 average rating

These categories consistently receive below-average customer ratings. Management should investigate product quality, supplier performance, packaging quality, and delivery conditions for these categories to improve customer satisfaction.

ROOT CAUSE HYPOTHESIS FOR LOW RATINGS

The presence of "Office Furniture" among the worst-rated categories suggests that bulk and dimensional weight play a role in dissatisfaction. Heavy, oversized items are more prone to shipping damage and complex last-mile delivery failures. Conversely, low ratings in "Security & Services" and "Diapers & Hygiene" likely point to product quality mismatches or counterfeit goods. Sellers in these specific categories should undergo a rigorous quality audit to protect the marketplace's reputation.

9. What Factors Contribute to Low Review Scores?

A strong relationship exists between delivery speed and customer satisfaction. Average Delivery Time by Review Score:

- **1-Star Reviews:** 19.12 days
- **2-Star Reviews:** 15.34 days
- **3-Star Reviews:** 13.58 days
- **4-Star Reviews:** 11.75 days
- **5-Star Reviews:** 10.19 days

Customers who waited longer for delivery consistently gave lower ratings. This finding suggests that delivery performance is one of the most influential factors affecting customer satisfaction. Reducing delivery times is likely to improve review scores and overall customer experience.

THE CRITICAL FULFILLMENT THRESHOLD

The data establishes a precise operational threshold: there is a **~9-day gap** between creating a brand promoter (5-star review at 10.19 days) and generating a detractor (1-star review at 19.12 days). This is the most actionable insight in the dataset. Management must implement strict Service Level Agreements (SLAs) targeting a maximum 12-day delivery window nationwide. Any order projected to exceed 14 days should automatically trigger proactive customer support outreach to mitigate the inevitable 1- or 2-star review.

Final Recommendations & Strategic Action Plan

1. **Leverage Southeast Density:** Optimize logistics and warehousing strictly around the São Paulo, Rio de Janeiro, and Minas Gerais corridor to secure the base generating the vast majority of the 15.92M BRL revenue.
2. **Overhaul Retention Mechanics:** Urgently address the 3.12% repeat customer rate by implementing loyalty incentives and aggressive post-purchase email marketing to reduce dependency on new customer acquisition.

3. **Establish the 12-Day SLA:** Since delivery times directly dictate review scores (10 days = 5 stars; 19 days = 1 star), reconfigure carrier contracts to guarantee a sub-12-day delivery standard.
4. **Audit High-Risk Categories:** Immediately pause or review sellers in the "Security & Services" and "Office Furniture" categories due to their highly toxic review averages, which drag down the platform's overall NPS.
5. **Solve the Northern Bottleneck:** Seek alternative regional courier partnerships for Roraima, Amapá, and Amazonas to combat the unacceptable 25+ day transit times.
6. **Capitalize on High-Margin Winners:** Redirect marketing budget to boost "Health & Beauty" and "Watches & Gifts" visibility, capitalizing on their smaller dimensional shipping profiles and strong sales volume.